GABAPENTIN

Respiratory depression with opioids

Controlled in some countries

CHEMICAL NAME 2-[1-(aminomethyl)cyclohexyl]acetic acid
CLASS Gabapentinoid (antiepileptic, neuropathic analgesic)
TRADE NAMES Neurontin · Gabarone · Gralise

THERAPEUTIC USES / INDICATIONS

- Neuropathic pain (diabetic neuropathy, post-herpetic neuralgia).
- Focal seizures (adjunct/monotherapy).
- Restless legs syndrome; pruritus (uraemic), hot flushes (off-label).

ADMINISTRATION

- **PO** three times daily (≤12 h between doses), with or without food.
- Separate from antacids by 2 h.
- Titrate upward over days; taper over ≥1 week to stop.

DOSAGES

- **Neuropathic pain:** day 1 300 mg once, day 2 300 mg twice, day 3 300 mg three times; increase to max 3.6 g/day.
- **Epilepsy:** same titration; usual 0.9–3.6 g/day.
- **Renal impairment:** CrCl 30–59: 400–1,400 mg/day; 15–29: 200–700 mg/day; <15: 100–300 mg/day.

PHARMACOKINETICS

- **Onset:** 1–3 h. Bioavailability saturable (↓ at high doses).
- Not metabolised; not protein-bound.
- Excreted unchanged in urine. **t½:** 5–7 h.

PHARMACODYNAMICS

- Binds **α2δ subunit of voltage-gated calcium channels** → ↓ release of excitatory neurotransmitters (glutamate, substance P).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dizziness, drowsiness, ataxia.
- Peripheral oedema, weight gain.
- Blurred vision, dry mouth.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression (with opioids, elderly, renal impairment, COPD).
- Misuse and dependence; withdrawal.
- Suicidal ideation; DRESS; pancreatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** renal impairment, respiratory disease, opioid co-prescription, history of substance misuse.

KEY INTERACTIONS

- **Opioids, benzodiazepines, alcohol:** additive CNS/respiratory depression.
- **Antacids (Al/Mg):** ↓ absorption by 20%.
- **Morphine:** ↑ gabapentin levels.

NURSING MANAGEMENT

- Calculate dose by renal function.
- Monitor sedation, RR (with opioids), falls risk.
- Assess pain scores and seizure frequency.
- Watch for misuse.

PATIENT EDUCATION

- May cause dizziness and sleepiness; do not drive until you know how it affects you.
- Do not stop suddenly.
- Avoid alcohol; tell doctor about opioid use.
- Report low mood or breathing difficulty.

KEY POINTS

- α2δ calcium-channel ligand.
- Renally excreted — adjust dose.
- Neuropathic pain; titrate from 300 mg.
- Respiratory depression with opioids; misuse.